

MSAI 699 Capstone Project

Capstone Project Proposal:

**NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence
Platform for Personalized Cognitive Health Education Using
Machine Learning and Retrieval-Augmented Generation**

Moriah Holland

University of the Cumberland

Submitted to the University of the Cumberland
in Partial Fulfillment of the Requirements of the Degree of
Master of Science in Artificial Intelligence

06/2026

Abstract

Artificial intelligence has transformed access to healthcare information through advances in natural language processing, machine learning, and large language models. However, general-purpose AI systems often generate inaccurate or unsupported medical information because they lack mechanisms for validating responses against reliable sources. These limitations are especially concerning within cognitive health education, where patients and caregivers require trustworthy, evidence-based information regarding Alzheimer's disease, mild cognitive impairment, genetics, and lifestyle interventions.

This project proposes **NeuroGuide AI**, an explainable multi-agent artificial intelligence platform that combines Retrieval-Augmented Generation (RAG), machine learning, explainable artificial intelligence (XAI), and specialized AI agents to deliver personalized cognitive health education. The system will retrieve and validate information from trusted medical sources while providing educational risk assessments using machine learning and feature-level explanations through SHAP.

The project will evaluate the platform using retrieval quality, response grounding, machine learning performance metrics, explainability, and usability measures. The goal is to demonstrate that integrating machine learning, Retrieval-Augmented Generation, explainable AI, and multi-agent architectures can improve the accuracy, transparency, and educational value of AI-assisted cognitive health education.

Proposal

Project Title

NeuroGuide AI: An Explainable Multi-Agent Artificial
Intelligence Platform for Personalized Cognitive Health Education
Using Machine Learning and Retrieval-Augmented Generation

Statement of the Project

Large language models have changed the way people access healthcare information by providing fast, conversational responses. Despite these advances, many AI systems remain susceptible to hallucinations, outdated knowledge, and limited transparency. These issues are particularly important in healthcare, where inaccurate information may negatively influence user understanding and decision-making.

The purpose of this project is to develop **NeuroGuide AI**, a multi-agent artificial intelligence platform designed to provide reliable, evidence-based cognitive health education. Unlike traditional chatbots, NeuroGuide AI combines Retrieval-Augmented Generation, machine learning, explainable AI, and multiple specialized AI agents to retrieve trusted medical literature, validate information, generate personalized educational responses, and explain machine learning predictions.

The platform will include a machine learning–based educational risk assessment that considers demographic, lifestyle, cardiovascular, and APOE $\epsilon 4$ genetic factors. SHAP explainability will

provide transparent feature-level explanations for every prediction, helping users better understand the factors influencing their educational risk assessment.

This project integrates multiple artificial intelligence disciplines studied throughout the Master of Science in Artificial Intelligence program, including machine learning, natural language processing, Retrieval-Augmented Generation, explainable AI, human-computer interaction, software engineering, cloud computing, and responsible AI. The completed system will be evaluated to determine whether a multi-agent architecture can improve the accuracy, transparency, and educational value of AI-assisted cognitive health education compared with traditional single-agent conversational AI systems.

Research Questions

Primary Research Question

Can an explainable multi-agent Retrieval-Augmented Generation (RAG) architecture provide more accurate, transparent, and trustworthy cognitive health education than a traditional single-agent conversational AI system?

Secondary Research Questions

RQ1: Does a multi-agent architecture improve the quality and reliability of educational responses compared to a single-agent approach?

RQ2: Can machine learning–based educational risk assessments improve personalization while maintaining transparency through explainable AI?

RQ3: How does Retrieval-Augmented Generation improve factual grounding and reduce unsupported responses?

RQ4: How do users perceive the usability, trustworthiness, and educational value of a multi-agent cognitive health platform?

Hypothesis

It is hypothesized that NeuroGuide AI will produce more accurate, transparent, and trustworthy educational responses than a traditional single-agent conversational AI system. By combining Retrieval-Augmented Generation, machine learning, explainable AI, and a multi-agent architecture, the platform is expected to improve factual grounding, reduce unsupported responses, and increase user confidence while providing personalized educational guidance.

Proposed AI Technologies

NeuroGuide AI integrates multiple artificial intelligence concepts explored throughout the Master of Science in Artificial Intelligence program.

Machine Learning

- Random Forest classification
- Educational risk prediction
- SHAP explainability

Natural Language Processing

- Large Language Models (GPT-4o-mini)
- TF-IDF document retrieval
- Latent Semantic Analysis (LSA)
- Prompt engineering

Retrieval-Augmented Generation

- Hybrid sparse and semantic document retrieval
- Citation-grounded response generation
- Confidence scoring

Multi-Agent Artificial Intelligence

The platform consists of four specialized AI agents that independently perform retrieval, reasoning, validation, and response synthesis.

Human-Computer Interaction

- Personalized user interface
- Patient and clinician response modes
- Accessibility-focused interface design

Responsible AI

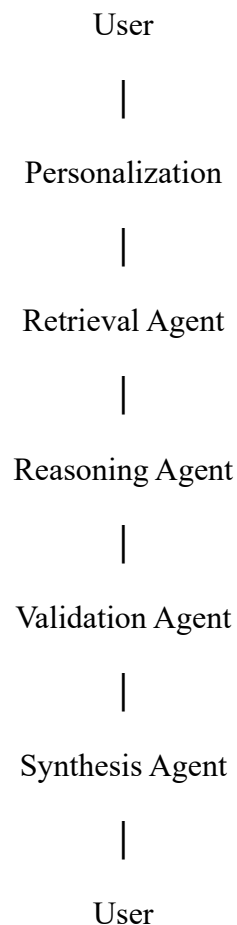
- Explainable AI (SHAP)

- Bias awareness
- Source transparency
- Educational medical disclaimers

Proposed System Architecture

NeuroGuide AI processes each user request through a coordinated sequence of specialized AI agents. Rather than relying on a single language model, each agent performs a specific task that contributes to the accuracy, transparency, and reliability of the final response.

Model architecture:



The proposed architecture consists of four collaborating AI agents.

Personalization

User profile information, including demographic and educational risk factors, is used to personalize educational responses and risk assessments.

Retrieval Agent

Searches trusted cognitive health resources and retrieves the most relevant supporting evidence.

Reasoning Agent

Analyzes retrieved information and generates evidence-based educational explanations tailored to the user's request.

Validation Agent

Verifies that generated responses are supported by retrieved evidence and assigns a confidence score to improve transparency.

Synthesis Agent

Produces the final educational response while adapting language for either patient or clinician audiences and providing supporting citations.

Datasets

The knowledge base will consist of trusted public sources related to Alzheimer's disease and cognitive health, including:

- PubMed Central
- NIH MedlinePlus
- Alzheimer's Association
- Cochrane Reviews
- World Health Organization (WHO)
- Centers for Disease Control and Prevention (CDC)

The educational risk assessment model is based on publicly available dementia risk factors summarized by the **2024 Lancet Commission**, incorporating demographic, lifestyle, cardiovascular, and APOE $\epsilon 4$ genetic information.

Evaluation

The effectiveness of NeuroGuide AI will be evaluated using quantitative and qualitative methods to assess system performance, machine learning accuracy, explainability, and user experience.

AI Performance

The Retrieval-Augmented Generation system will be evaluated using:

- Retrieval relevance
- Citation grounding
- Response confidence
- Response latency

Machine Learning

The educational risk assessment model will be evaluated using standard machine learning metrics, including:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Explainable AI

Model transparency will be assessed using SHAP feature attribution to verify that educational risk predictions are explainable and clinically reasonable.

Human-Computer Interaction

User experience will be evaluated through:

- System Usability Scale (SUS)
- User satisfaction
- Accessibility review
- Response readability

These evaluation methods will determine whether the proposed multi-agent architecture improves the reliability, transparency, and educational value of AI-assisted cognitive health education.

Ethical Considerations

Because NeuroGuide AI operates within a healthcare education domain, ethical AI practices are essential throughout the project.

The platform will function solely as an educational resource and will not provide medical diagnoses or treatment recommendations. Every response will include a disclaimer encouraging users to consult qualified healthcare professionals for medical advice.

To improve transparency, all responses will include citations from trusted medical sources whenever possible. Explainable AI techniques using SHAP will allow users to understand how individual risk factors contribute to educational risk assessments.

The project will not collect or permanently store protected health information. Potential sources of bias will be considered throughout development by using evidence-based medical resources and evaluating responses for fairness and consistency.

Demonstration of Program Learning Outcomes

Program Learning Outcome 1

Apply artificial intelligence principles to solve complex real-world problems.

This project applies machine learning, Retrieval-Augmented Generation, natural language processing, and multi-agent artificial intelligence to address the challenge of providing accurate and trustworthy cognitive health education.

Program Learning Outcome 2

Demonstrate independent research skills using artificial intelligence literature.

The project integrates peer-reviewed research, trusted healthcare resources, and current artificial intelligence methodologies to support system development and evaluation.

Program Learning Outcome 3

Develop and evaluate artificial intelligence solutions.

NeuroGuide AI will be designed, implemented, and evaluated using established performance metrics, usability testing, and explainability techniques to assess its effectiveness as an educational platform.

Program Learning Outcome 4

Evaluate ethical, legal, and societal implications of artificial intelligence.

The project incorporates responsible AI principles through transparency, explainability, source attribution, educational disclaimers, and consideration of bias within healthcare applications.

Program Learning Outcome 5

Communicate artificial intelligence concepts effectively.

The project will include technical documentation, system architecture, evaluation results, and a formal research paper that communicate complex artificial intelligence concepts to both technical and non-technical audiences.

Conclusion

NeuroGuide AI demonstrates how multiple artificial intelligence techniques can be integrated into a single educational platform that promotes trustworthy, evidence-based cognitive health information. By combining machine learning, Retrieval-Augmented Generation, explainable AI, and a multi-agent architecture, the project seeks to improve the accuracy, transparency, and personalization of AI-assisted healthcare education while showcasing the knowledge and skills developed throughout the Master of Science in Artificial Intelligence program.